

Research Article

Application of a GIS-Based DRASTIC Model for Groundwater Vulnerability Assessment: A Case Study of Lai City, Tandjile East Department, Southern Chad

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Abstract

Groundwater resource protection is a worldwide issue because groundwater plays a crucial role in meeting water supply demands for both natural ecosystems and human activities. Effective groundwater management and protection therefore require appropriate approaches for assessing contamination risks and identifying areas that are particularly vulnerable to pollution. In this study, a GIS-based DRASTIC model was applied to assess groundwater vulnerability in Lai City, southern Chad. Seven hydrogeological parameters were considered, namely depth to water, net recharge, aquifer media, soil media, topography, impact of the vadose zone, and hydraulic conductivity. A weighted overlay analysis based on the DRASTIC Index (DI) was subsequently used to generate the groundwater vulnerability map of the study area. The results show that the DRASTIC Index values range from 119 to 175, enabling the study area to be categorized into three groundwater vulnerability classes: low, moderate, and high. The high-vulnerability class covers approximately 35% of the study area, highlighting areas that require particular attention for groundwater protection and management. The average and low class of vulnerability covers portions of the study area 31% and 34%, respectively. The northern and south-western portion of the catchment, namely Lai, Gabri Mbassa, Sategui, and Django, possesses high groundwater vulnerability. GIS-based Drastic model validation is performed via water quality parameters such as Nitrate and Total Dissolved Solids, with the accuracy of 97% and 75% respectively. The resulting groundwater vulnerability map provides a valuable tool for groundwater protection and sustainable land-use planning in Lai City and its surrounding areas.

Keywords

Drastic, Geographic Information System, Groundwater Vulnerability, Nitrate and Total Dissolved Solids Concentration, Lai City

1. Introduction

Groundwater resources are the most precious source of fresh water available from natural resources on the earth [13, 49]. It is the main source of drinking water supply in many countries because of its availability, easy accessibility, and

low vulnerability to pollution compared to surface water resources [29]. According to data from Global Water Partnership (GWP), approximately 20% of Chad's groundwater withdrawals are allocated to human consumption, while the

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remaining 80% are used for agricultural purposes [27]. Despite their importance in human life in most parts of the world, aquifers have been highly exposed to pollution in recent years, and this situation limits the potential of groundwater resources for use [20, 33, 62]. Groundwater contaminant sources could be natural (e.g., natural deposits of gypsum, minerals, salts, seawater intrusion) or anthropogenic (untreated livestock waste, nitrates, pesticides.). Among the various sources of groundwater pollution, anthropogenic pollution represents one of the most significant threats to groundwater resources and is increasingly recognized as a major contributor to groundwater quality degradation, particularly in urban environments [32, 66, 71]. Groundwater quality in the southern part of the Lake Chad basin, which includes the town of Lai, has shown that anthropogenic activities contribute to the degradation of groundwater quality [12, 64]. The protection and restoration of groundwater resources have become increasingly important in recent years, highlighting the need to assess groundwater contamination in areas that may be potentially affected. However, the groundwater vulnerability assessment is needed in this context for taking remedial steps.

Aquifer vulnerability concept is introduced first by Margat (1968) [37]. This concept refers to the capacity of contaminants to diffuse and infiltrate from the soil surface into the underlying aquifer system. The approach has been successfully applied under a wide range of climatic conditions worldwide, including arid and semi-arid environments [10, 39, 73], tropical and subtropical regions [26, 59, 63], and temperate climates [36, 41]. Furthermore, it is implemented across a wide range of hydrogeological settings, including coastal regions [21, 43], Hard rock aquifers [51, 60], Karst aquifers [11, 45, 70], Alluvial aquifers [6, 24]. Vrba and Zaporozec (1994) [72] described groundwater vulnerability as an intrinsic property of an aquifer system that reflects its susceptibility to natural and anthropogenic impacts. This vulnerability is governed by various intrinsic characteristics of the aquifer and its surrounding environment, including soil properties such as structure and texture, as well as hydrological and topographic factors such as drainage density, runoff volume, and slope [2, 28, 54]. Therefore, a groundwater vulnerability assessment approach in a given aquifer system is put in place to delineating areas likely to be impacted by contaminants derived from anthropogenic activities.

In recent years, aquifer vulnerability assessment has emerged as an essential tool for preventing groundwater pollution. However, several approaches can be used to assess groundwater vulnerability such as: process-based model [46, 74], DRASTIC [7], GOD [23], AVI [69], SINTACS [16], EPIK [19], SIGA [72], PCA technique [53], Decision random forest [34], tree-based data mining [77], Fuzzy Clustering [31], Boosted regression tree [44]. The selection of a suitable groundwater vulnerability model depends largely on the type of aquifer, the availability of relevant data, and the specific objectives and scope of the study [13, 20]. Among the various

groundwater vulnerability assessment methods, the DRASTIC model—based on seven hydrogeological parameters, namely depth to groundwater (D), net recharge (R), aquifer media (A), soil media (S), topography (T), impact of the vadose zone (I), and hydraulic conductivity (C)—is one of the most widely used approach. Its applicability has been demonstrated in numerous studies [4, 5, 13, 20, 42, 61, 63, 67]. The DRASTIC method is based on four fundamental assumptions [7, 20]: contaminants are introduced at the ground surface; precipitation transports them toward the groundwater; their mobility is assumed to be equivalent to that of water; and the assessment area is generally expected to cover at least 0.4 km². The GIS-based DRASTIC model, which integrates the DRASTIC framework with GIS-based spatial analysis and mapping techniques, has been widely applied to delineate and map groundwater vulnerability zones [13, 20]. This integrated approach offers a cost-effective and time-efficient means of assessing and mapping groundwater vulnerability.

Groundwater represents the main resource for domestic needs in Lai city (Figure 1). The problem we are currently facing stems from the fact that the study area is a floodplain where agriculture is the main activity of the population. Agricultural activities are degrading the groundwater quality of Lai city due to the uncontrolled use of fertilizers and pesticides. Furthermore, field observations indicate that the city is facing rapid urbanization, accompanied by inadequate sanitation infrastructure, as evidenced by the widespread use of open latrines and the lack of adequate wastewater and sewage disposal facilities. It appears essential to conduct a study to highlight the parameters and conditions that control the pollution of the groundwater in the Lai city. However, few studies exist to date regarding groundwater pollution and the assessment of groundwater vulnerability through GIS-based DRASTIC vulnerability model in the Lai area although research on this issue has been conducted at the scale of the Lake Chad Basin [14, 40, 47], of which the study area is a part. In this context, groundwater vulnerability assessment of Lai city is needed in order to carefully predict pollution that faced by the city and its surroundings. The ultimate objective of this study is to evaluate the groundwater vulnerability of Lai City by integrating the DRASTIC model into GIS and also discuss the spatial distribution of nitrate concentration in the groundwater. Ultimately, this study provides a useful basis for planners and local authorities to develop an effective groundwater management and protection plan for Lai City. This research was carried out in N'Djamena, Chad, throughout the period of March 2024 to January 2025.

2. Materials and Methods

2.1. Study Site

Tandjile is a region of southern Chad where Lai City is the capital. The study area, corresponding to Lai City and illus-

trated in Figure 1, covers approximately 151.12 km². Geographically, it extends from 8°31'20" N to 9°20'30" N in latitude and from 16°13'40" E to 16°22'20" E in longitude. The hydrographic network of the study area comprises three main rivers: the Logone, Tandjilé and Kabia Rivers. The Logone River is the principal river in the study area. It originates in the highlands of central Cameroon and flows northward toward Lake Chad. The local economy is primarily supported by agricultural and fishing activities, supplemented by livestock farming. Meteorological records, including precipitation and temperature data obtained from the National Agency of Meteorology for the period 1991–2023, indicate that the study area receives an average annual rainfall of approximately 1,008 mm. The rainy season extends for up to seven months, from April to October. The mean annual temperature varies

between 27.1 °C and 29.2 °C, while the elevation of the study area ranges from 353 to 374 m above sea level. The geological formations of the Tandjilé region, of which the study area is a part, are represented by two large groups characteristic of sedimentary zones: the Tertiary, represented by the Continental Terminal formation, and the Quaternary, represented by fluvial-lacustrine deposits [58]. The hydrogeological context is characterized by the existence of water tables in continuous sedimentary formations (free, semi-captive or captive) and in discontinuous basement formations [57]. In fact, continuous sedimentary formations are mostly located in fluvial-lacustrine layers with free water tables. However, in the extreme south-eastern part of the study area, the roof of the water table is impermeable in places [58].

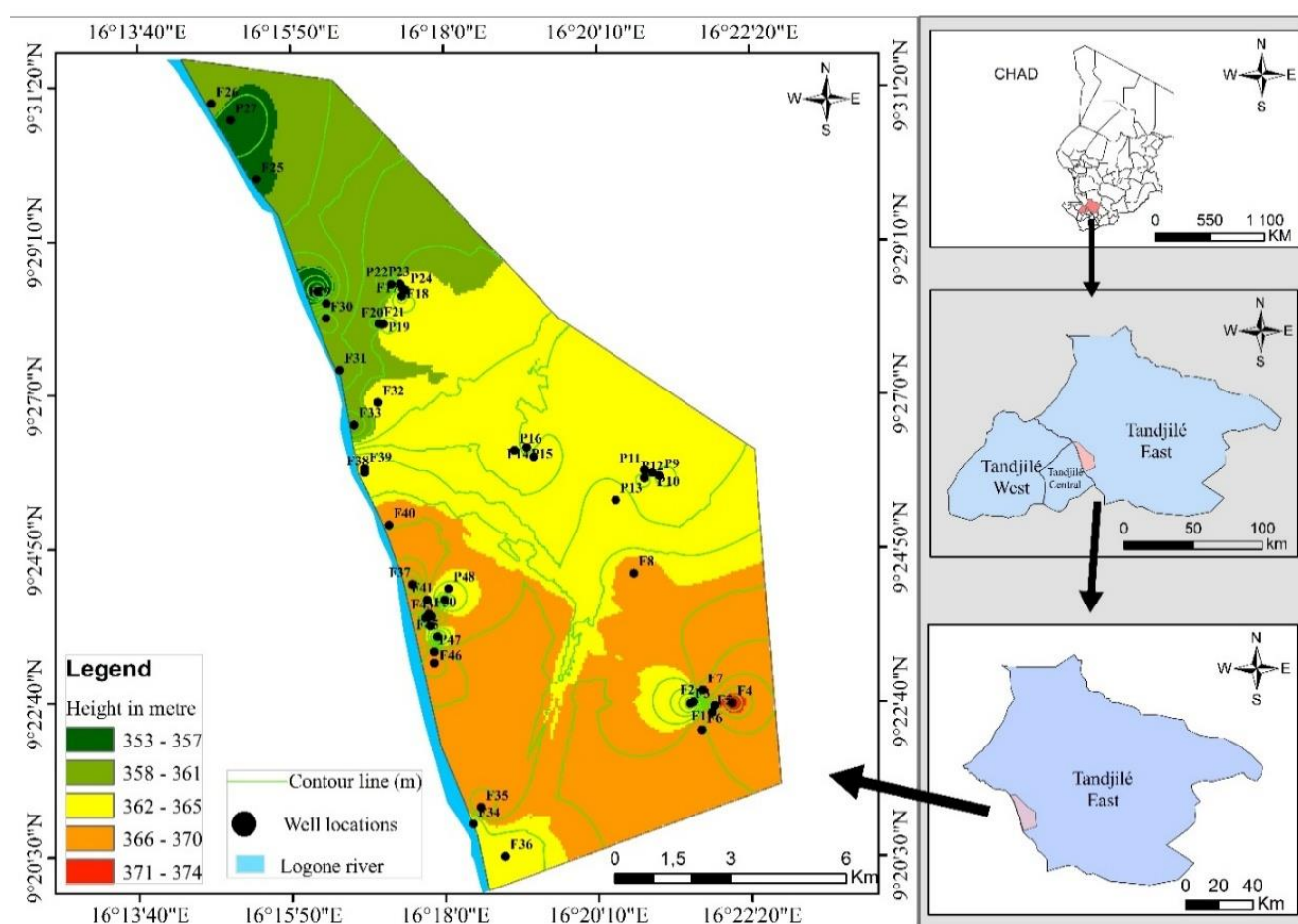


Figure 1. Study Area Location.

2.2. Groundwater Vulnerability Assessment Using the DRASTIC Model

The DRASTIC model, originally developed by the U.S. Environmental Protection Agency (EPA), is among the most widely applied methods for assessing aquifer vulnerability to

contamination. The model evaluates groundwater susceptibility by integrating seven hydrogeological parameters through a weighted-index approach [7]: depth to water (D), net recharge (R), aquifer media (A), soil media (S), topography (T), impact of the vadose zone (I), and hydraulic conductivity (C). The Drastic approach is divided into three significant parts [7]: weights, ranges and rating. According to Aller et al. (1987) [7],

each DRASTIC parameter is assigned a relative weight ranging from 1 to 5, reflecting its relative significance in controlling groundwater flow and contaminant transport (Table 1). The weights of Drastic parameter are constant and may not be changed. The most significant parameter has weight of 5 (Depth to water level, vadose media) and least significant has a weight of 1 (topography). The parameter ratings associated to each Drastic factors are variable and user can define and calibrate values to suit a given region [18]. Note that, the parameter D, R, S, T, and C have been assigned one value per range. However, A and I have a “typical” and variable rating. The Delphi technique [7] has been used to establish numerical weight and rating. This method is well-known and used worldwide [9, 13, 52, 55, 61]. In this technique, practical and

research experiences of professionals in the area of interest are used to assess levels of risk [17, 52]. The DRASTIC index is determined by multiplying the rating assigned to each hydrogeological parameter by its corresponding weight, as presented in Table 2. Based on these data, thematic maps were generated for each hydrogeological parameter using a spatial resolution of 30 m × 30 m and subsequently reclassified according to their respective ratings and weights. The final groundwater vulnerability map was subsequently produced through a weighted-sum overlay analysis based on the DRASTIC index, following the methodological workflow shown in Figure 2.

DRASTIC Index (DI) could be expressed as [7]:

$$DI = DrDw + RrRw + ArAw + SrSw + TrTw + IrIw + CrCw \quad (1)$$

where D, R, A, S, T, I, C represent the seven hydrogeological parameters; *r* denotes the rating assigned to each parameter, while *w* represents its corresponding weight.

A higher DRASTIC Index (DI) values correspond to greater

groundwater vulnerability and consequently a higher potential for contamination, whereas lower DI values reflects a lower vulnerability [13].

Table 1. Assigned weights of the Drastic parameters [7].

Parameter		Generic weight
Depth to water	D	5
Net recharge	R	4
Aquifer media	A	3
Soil media	S	2
Topography	T	1
Impact of vadose zone	I	5
Hydraulic conductivity	C	3

Table 2. DRASTIC Parameter Ratings and Weights for the Study Area.

Parameters	Range	Rating	Weight	Total weight (rating x weight)
Depth to water level (in meter)	1.5 - 4.5	9	5	45
	4.5 - 9	7		35
Net recharge (mm/year)	118	6	4	24
	Compact clay	4		12
	Lateritic clay	5		15
Aquifer media	Sandy clay	6	3	18
	Fine sand	8		24
	Medium sand	9		27
Soil media	Tropical ferruginous soil	2	2	4

Parameters	Range	Rating	Weight	Total weight (rating x weight)
Topography (slope in%)	Holomorphic soil	4	1	8
	Hydromorphic soil	8		16
	0 - 1	5		5
	2 - 3	7		7
	4 - 9	9		9
Vadose zone media	Clay or silt	3	5	15
	Sandy clay	6		30
	Silty sand	8		40
Hydraulic conductivity (m/s)	Sand	9	3	45
	5×10^{-4}	8		24

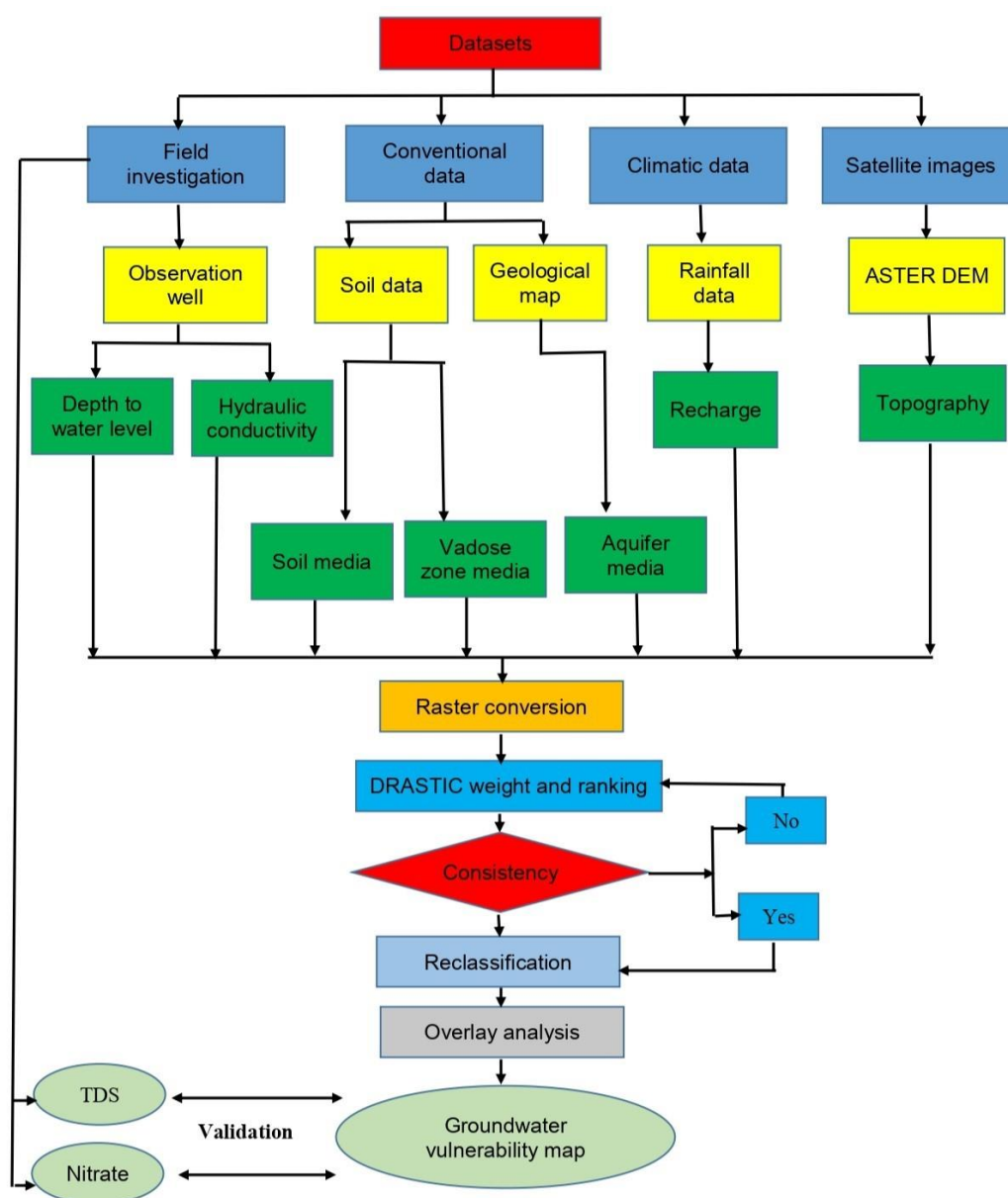


Figure 2. Workflow for Groundwater Vulnerability Assessment and Mapping.

2.3. Data Collection and Analysis

To apply the DRASTIC model previously described, seven parameters are necessary. These parameters are: depth to water level, net recharge, aquifer media, soil media, topography, impact of vadose zone, and hydraulic conductivity. The data relating to the depth of the water level were collected from a field survey conducted in April 2024. A total of 50 well locations have been selected in the study area for measuring depth to water level and main groundwater quality parameters, including Nitrate and TDS. Groundwater samples were analyzed for selected quality parameters, including nitrate and total dissolved solids (TDS), at the LHGR Laboratory of the University of N'Djamena. The analyses were conducted in accordance with the standard methods recommended by the American Public Health Association [8]. The spatial distribution map of Depth to water level, TDS, and Nitrate was prepared using the standard kriging tool available in ArcGIS 10.8.1. Rainfall data has been collected from the National Agency of Meteorology. The annual rainfall values were used to calculate the recharge using a simple water balance model [22]. The data relating to aquifer media are obtained from well log data available in the CDIG (Geographical Information Centre of the Water Ministry of CHAD) archives. The soil media map is obtained by digitalization of the pedological map produced by Pias (1970) [50] in the study area. ASTER DEM (30 m spatial resolution) has been used for generating the topography of the study area using the surface tools of ArcGIS 10.8.1. The data relating to the impact of the vadose zone and hydraulic conductivity were obtained based on the data collected from CDIG.

2.4. Validation of Groundwater Vulnerability Map

Validation of the vulnerability map derived from the DRASTIC model is needed before the use of its results. For this reason, a validation procedure was adopted to evaluate the accuracy of the DRASTIC model. Contaminant data sets collected from wells throughout the study area can provide an effective basis for validating the groundwater vulnerability map [20]. Nitrate and total dissolved solids (TDS), which are common indicators of groundwater contamination, were selected as validation parameters for the groundwater vulnerability assessment, consistent with the methodology proposed by Bera et al. (2021) [13]. Along the same lines, Elmeknassi et al. (2021) [20] used groundwater contaminant parameters such as nitrate concentrations and electrical conductivity measured in the study site to validate the vulnerability map. Firstly, a superimposition of nitrate and TDS data on the vulnerability map was carried out to obtain a qualitative validation of the model. High values of the studied chemical compounds (nitrate and TDS) should be found in the high vulnerability zone, and conversely, low concentrations in the low vulnerability zone. This validation procedure was adopted by several authors [13, 20, 76].

Secondly, the relationship between the DRASTIC Index (DI) and nitrate and TDS concentrations was further assessed using the Pearson correlation coefficient (r). This statistical measure is commonly used to quantify the strength and direction of the linear association between two continuous variables (Singh et al., 2015). The Pearson correlation coefficient (r) was calculated using the following equation [13]:

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}} \quad (2)$$

where r is the Pearson correlation coefficient, n represents the number of pairs, and x and y denote the two variables considered.

At the end, the maximum nitrate and TDS concentrations were determined for each groundwater vulnerability class and compared with the corresponding DRASTIC Index (DI) values. A similar procedure was adopted by El-Meknassi et al. (2021) [20].

3. Result and discussion

3.1. Depth to Water Level

The first DRASTIC parameter, depth to water (D) level, is defined as the vertical distance between the ground surface and the water table [76]. This hydrogeological parameter is a key component of the DRASTIC model because the unsaturated zone between the ground surface and the water table provides an important pathway for contaminant attenuation before pollutants reach the groundwater. In general, greater depth to the water table provides a longer travel distance and greater opportunity for contaminant attenuation, thereby reducing groundwater vulnerability to contamination [7]. Water level measurements are carried out in April 2024 in order to perform a depth to water level map. A total of 50 water points, comprising 15 wells and 35 boreholes, were surveyed. These wells and boreholes are located in the study area (Lai City) and its surroundings and used for domestic needs. The locations and measured water levels of the 50 wells and boreholes were used to interpolate the depth to the water table using the ordinary kriging method available in the Spatial Analyst extension of ArcGIS 10.8.1. The depth to the water table across the study area ranges from 1.5 to 9.0 m. Based on the DRASTIC classification scheme, these values were divided into two classes: 1.5–4.5 m and 4.5–9.0 m, which were assigned ratings of 45 and 35, respectively (Table 1). The southwestern and northwestern parts of the study area, including Lai, Sategui, and Gabri Mbassa, are characterized by relatively shallow groundwater depths ranging from 1.5 to 4.5 m. These areas are therefore considered more vulnerable to groundwater contamination. In contrast, the southeastern and northeastern sectors, including Djoun, Totalie, and Madjang, exhibit greater depths

to the water table, ranging from 4.5 to 9.0 m, indicating comparatively lower groundwater vulnerability to contamination.

The resulting depth-to-water map is presented in Figure 3.

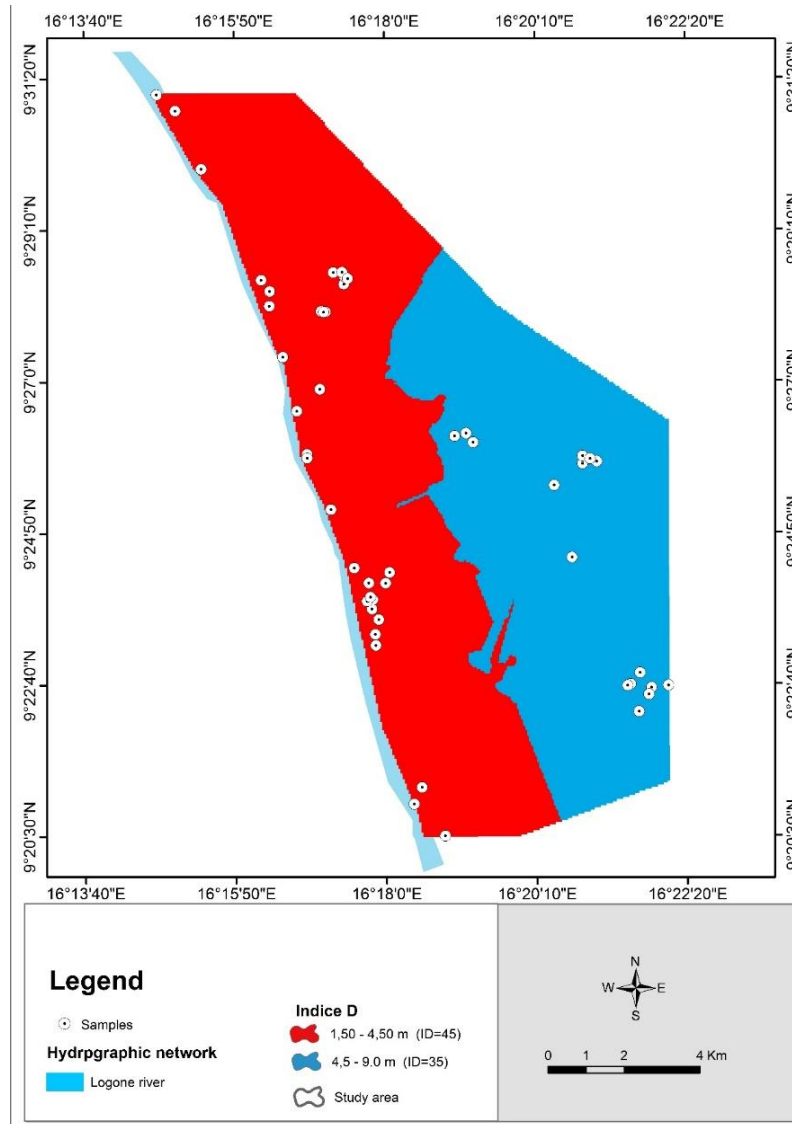


Figure 3. Depth to water level map.

3.2 Net Recharge

Net recharge refers to the amount of water that infiltrates through the soil and ultimately reaches the groundwater [7]. It is influenced by several factors, including slope, soil permeability, precipitation, land cover, and infiltration rate [7]. Higher recharge rates generally increase the potential for contaminants to be transported downward and reach the groundwater table, whereas lower recharge rates tend to reduce this potential. In this study, net recharge was estimated using meteorological data covering the period from 1990-1991 to 2022-2023, based on the classical water-balance approach and the following equation:

$$NR = P - ET - R_o \quad (3)$$

where, NR: is the net recharge in mm/year, P: is the annual precipitation in mm; ET is evapotranspiration in mm/year, R_o is the total runoff in mm.

The average total yearly precipitation P calculated which is about (1008) mm/year. ET was calculated based on the well-known Thornthwaite Method (1948) [68]. R_o was evaluated using the Soil Conservation Service (SCS) method (SCS-CN) to estimate the total runoff for the basin. The SCS-CN is the most suitable method [49] and has been applied by several authors [1, 3, 30]. First, the basin is divided into several curve number (CN) and then using the following equation:

$$Q = \frac{(P-0.2S)^2}{P+0.8S} \text{ for } P > 0.2S \quad (4)$$

$$S = \left(\frac{25,400}{CN} \right) - 254 \quad (5)$$

where: Q = accumulated runoff excess in (mm). P = accumulated average monthly rainfall (mm).

The annual runoff and the average net recharge calculated in the study area are 64 mm and 118 mm/year, respectively. The estimated net recharge values fall within a single class, ranging from 100 to 180 mm/year, which was assigned a total weight of 24 (Table 1). Based on these recharge data, a net recharge map was generated for the study area, as presented in Figure 4.

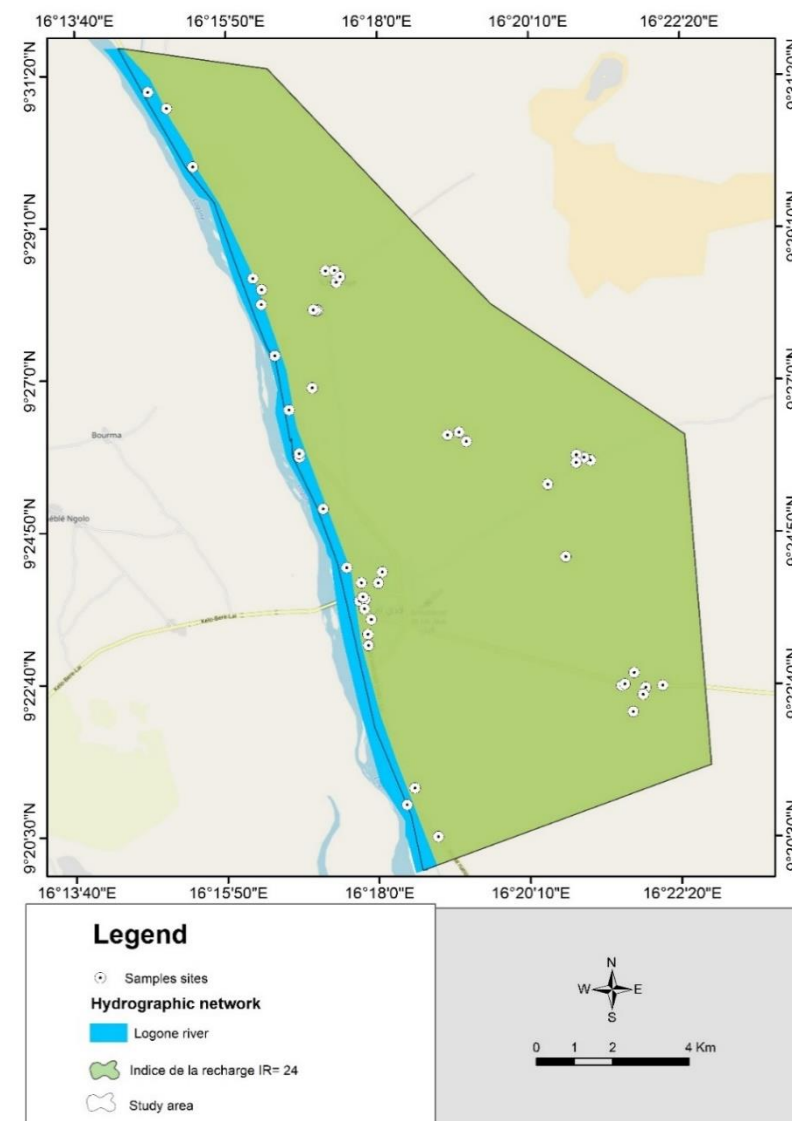


Figure 4. Net recharge map.

3.3. Aquifer Media

Aquifer media represent the porous and/or fractured geological formations that provide pathways for groundwater flow. The characteristics of the aquifer medium strongly influence contaminant transport and attenuation, as various physicochemical processes, including sorption, cation exchange, filtration, and dissolution, may occur during ground-

water movement. Therefore, aquifer media represents an important hydrogeological parameter in groundwater vulnerability and quality assessment. According to Lathamani et al. (2015) [35], larger grain sizes and a greater abundance of fractures, joints, or other openings enhance aquifer permeability, facilitating the migration of contaminants and consequently increasing groundwater vulnerability to contamination. The DRASTIC parameter A (aquifer media) was characterized using lithological data obtained from well logs available in the archives of the CDIG (Geographical Information Centre of the

Ministry of Water of Chad). These data indicate the presence of five main lithological formations within the study area: compact clay, lateritic clay, sandy clay, fine sand, and medium sand. Based on their relative influence on groundwater vulnerability, DRASTIC ratings were assigned to these aquifer media as follows (Table 1): compact clay was assigned a rating of 12, lateritic clay 15, sandy clay 18, fine sand 24, and medium sand 27. The resulting aquifer media map is presented in

Figure 5b.

In order to strengthen the aquifer media map interpretation, the lithological cross-sections for some water points (Tchindaye, Totalie, Djoun) has been included in the Figure 5(a), 5(c) and 5(d). These data obtained from CDIG archives confirm soils distribution in the study site.

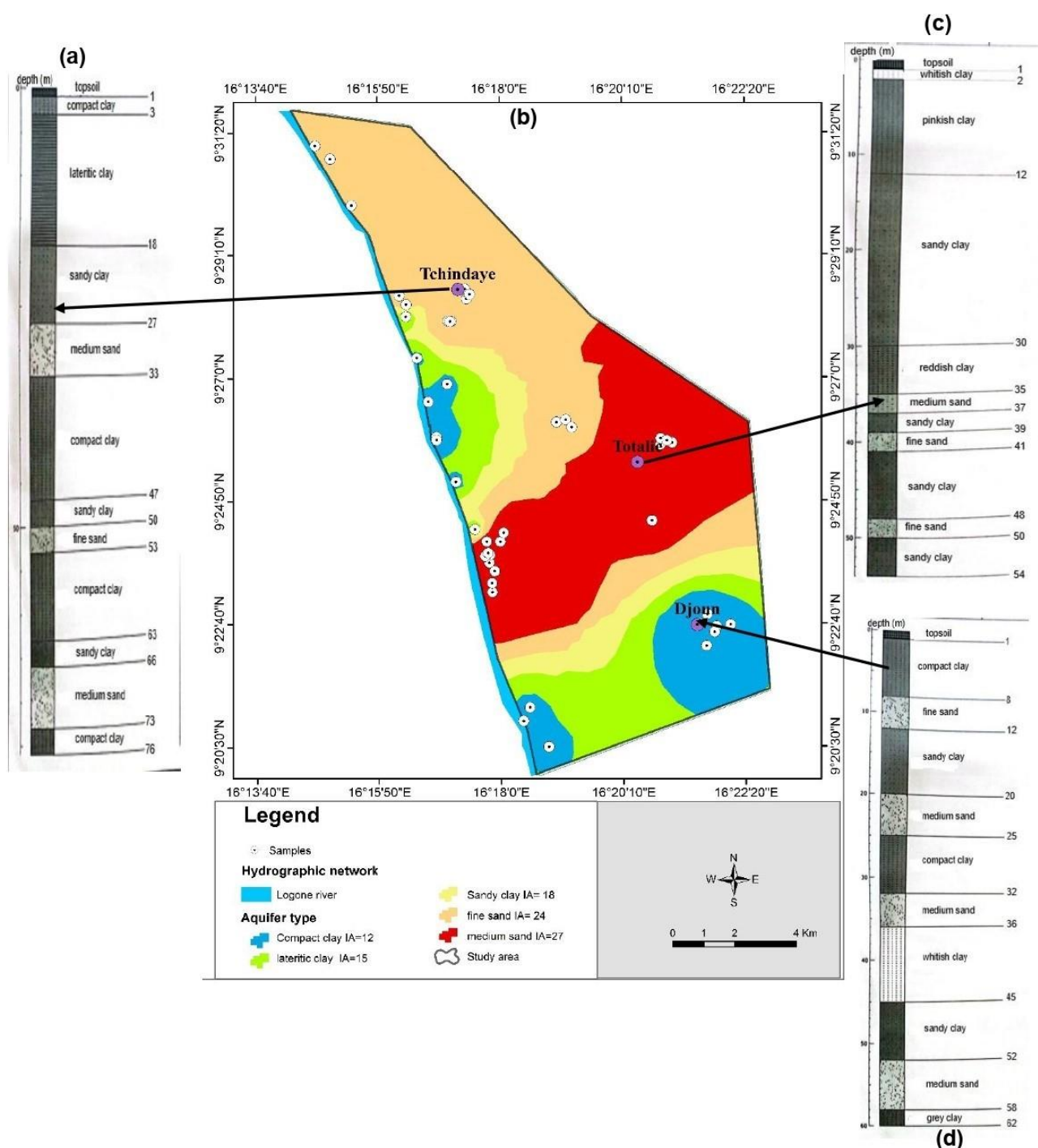


Figure 5. (b) Aquifer media map, (a) lithological cross-section for Tchindaye, (c) lithological cross-section for Djoun, (d) lithological cross-section for Totalie.

3.4. Soil Media

Soil media corresponds to the uppermost part of the vadose zone, where biological activity is particularly active [13]. The Drastic Factor S (soil media) plays an important role in the control of recharge processes and the attenuation and removal of contaminants during their migration through the soil profile. The soil media map of the study site (Figure 6) was developed by digitizing the pedological map produced by Pias (1970) [50]. Three main soil types were identified within the study area: tropical ferruginous soil, halomorphic soil, and hydromorphic soil. Based on their relative influence on groundwater vulnerability, DRASTIC ratings were assigned to the different soil media as follows (Table 1): tropical ferruginous soil was

assigned a rating of 4, halomorphic soil a rating of 8, and hydromorphic soil a rating of 16. Halomorphic soils are predominantly distributed in the southern part of the study area, particularly around Gabri-Bassa and Djoun, whereas hydromorphic soils occur mainly in the northern sector, particularly around Sategui. The presence of hydromorphic soils in the northern part of the study area may favor enhanced infiltration and contaminant transport, potentially increasing groundwater vulnerability in this sector. In the Figure 6 below, the photo of the representative soils types in the study site (hydromorphic soil (a'), tropical ferruginous soil (c'), halomorphic soil (d')) has been illustrated in order to provide valuable visual images and improve interpretation of soil characteristics in place at the site studied.

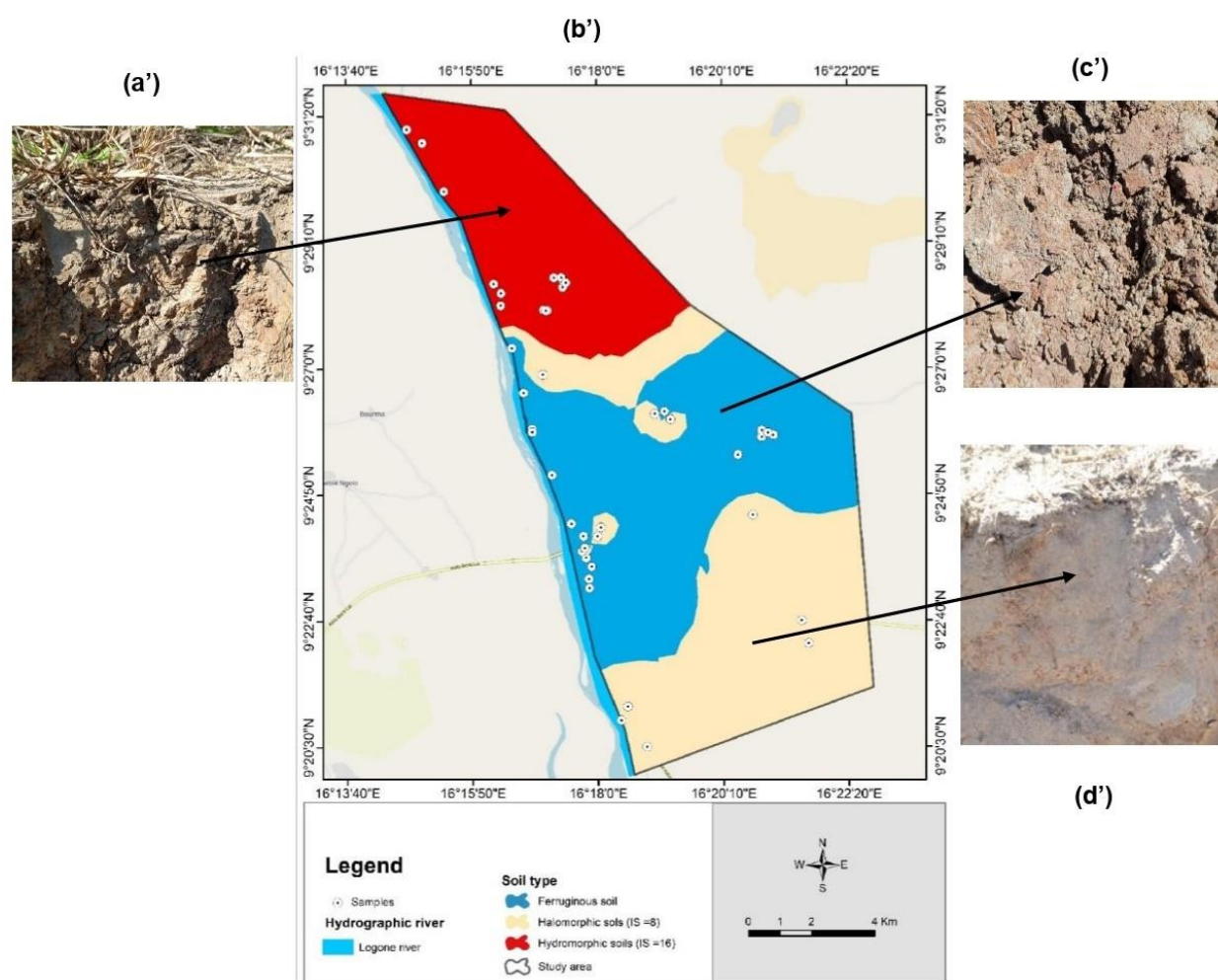


Figure 6. (a') hydromorphic soil, (b') soil media map, (c') tropical ferruginous soil, (d') halomorphic soil.

3.5. Topography

Topography, represented by slope, is an important parameter that impact surface runoff and contaminant infiltration [7]. Steep slopes generally promote rapid surface runoff, thereby

reducing the opportunity for contaminants to infiltrate into the subsurface. Conversely, gentle slopes favor water retention and infiltration, increasing the potential for contaminants to reach the groundwater system. The slope map of the study area (Figure 7) was derived from an Digital Elevation Model

(DEM) with a spatial resolution of 30 m. The DEM was processed using the 3D Analyst extension of ArcGIS 10.8.1 to generate the slope layer. Slope values across the study area range from 0 to 9% and were classified into three categories: 0-2%, 2-4%, and 4-9%. Each slope class was assigned a DRASTIC rating according to its relative influence on groundwater vulnerability, following the classification proposed by Aller et al. (1987) [7] (Table 1). The resulting slope

map indicates that the study area is predominantly flat, with most of the area characterized by slopes ranging from 0 to 2%. Such low-gradient conditions favor infiltration and limit surface runoff, thereby increasing the potential for contaminants to migrate toward the groundwater. Consequently, the predominantly gentle topography contributes to the relatively high groundwater vulnerability of the study area.

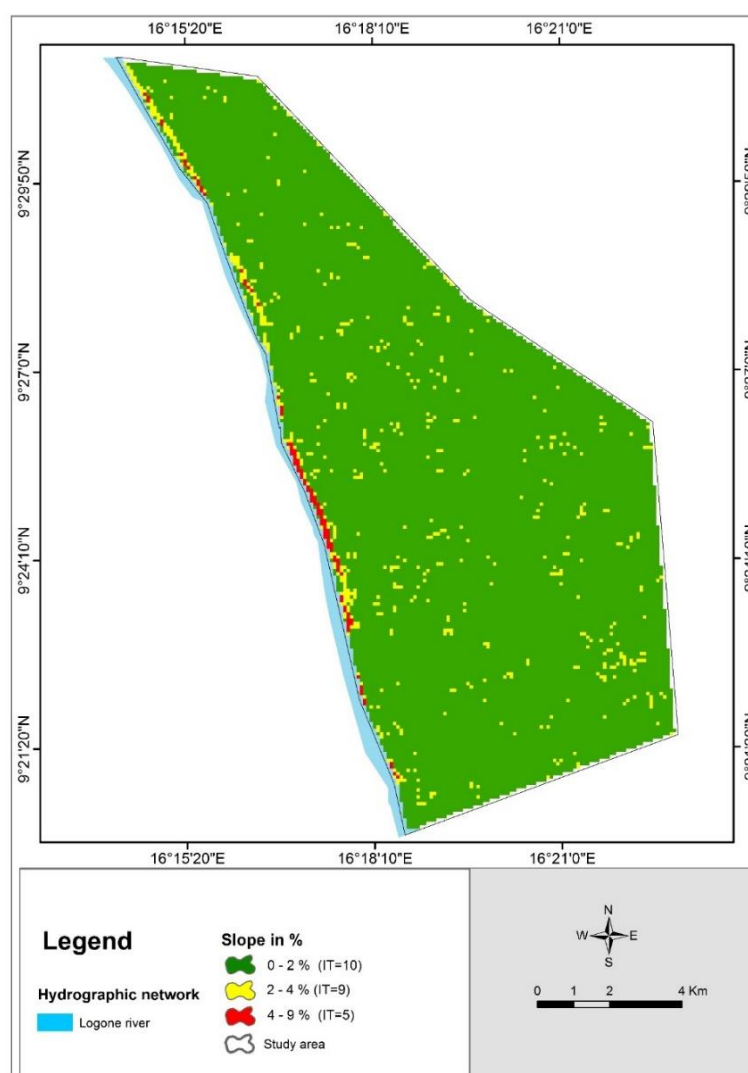


Figure 7. Topography map.

3.6. Impact of Vadose Zone

The unsaturated zone, referred to as the vadose zone, is the subsurface layer extending from the ground surface to the water table and overlying the saturated zone. This critical zone plays a role in controlling the downward transport and percolation of contaminant-laden water toward the groundwater system [75]. During contaminant migration through the vadose zone, several physicochemical and biological processes,

including mechanical filtration, biodegradation, dispersion, and volatilization, may occur. These processes contribute to the attenuation and transformation of contaminants before they reach the water table [7]. The vadose zone acts as a natural protective layer for groundwater, with its capacity to attenuate contaminants largely depending on its lithological characteristics. Data on vadose zone media were obtained from well logs available in the CDIG archives, using the same approach as that applied to characterize the aquifer media. The results indicate that the vadose zone in the study area consists

primarily of clay or silt, sandy clay, silty sand, and sand. A high DRASTIC rating of 45 was assigned to sand, which is predominantly distributed in the extreme northern part of the study area (Sategui) and in the central-eastern sector (Lai). Because of its relatively high permeability, sandy material facilitates contaminant migration and is therefore associated with

greater groundwater vulnerability. In contrast, clay occurring in the southeastern part of the study area, particularly around Djoun, was assigned a lower rating of 15, reflecting its lower permeability and greater capacity to impede contaminant transport. The resulting vadose zone media map is presented in Figure 8.

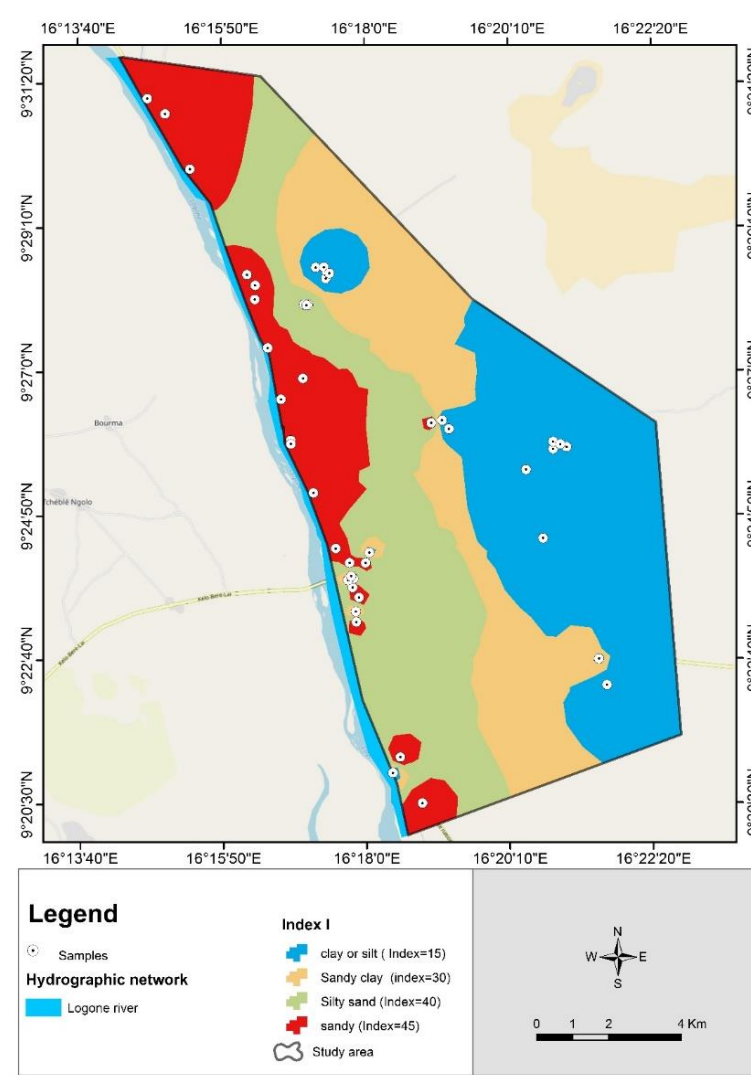


Figure 8. Vadose zone map.

3.7. Hydraulic Conductivity

Hydraulic conductivity is an important hydrogeological parameter that controls the rate and extent of contaminant transport within the groundwater system. Higher hydraulic conductivity values generally facilitate more rapid groundwater flow and contaminant migration, thereby increasing the potential for groundwater contamination [52]. It is more challenging to achieve an accurate estimation of hydraulic conductivity, and it is also considered a limitation of the DRASTIC approach [38, 65]. The hydraulic conductivity map of the

study area (Figure 9) was developed based on hydraulic conductivity values associated with the main rock types, as reported by Gilbert (1982) [25]. Based on the reference values provided in Gilbert (1982) [25], conductivity values were assigned to each aquifer media (compact clay: 0.2 m/day; lateritic clay: 11.15 m/day; sandy clay: 24.11 m/day; fine sandy clay: 38.97 m/day; medium sand: 43.2 m/day). The weighted scores assigned to the different hydraulic conductivity values corresponding to the aquifer media in the studied zone are presented as follows: 3 for compact clay, 9 for lateritic clay, 15 for sandy clay, 21 for fine sandy clay, and 24 for medium sand.

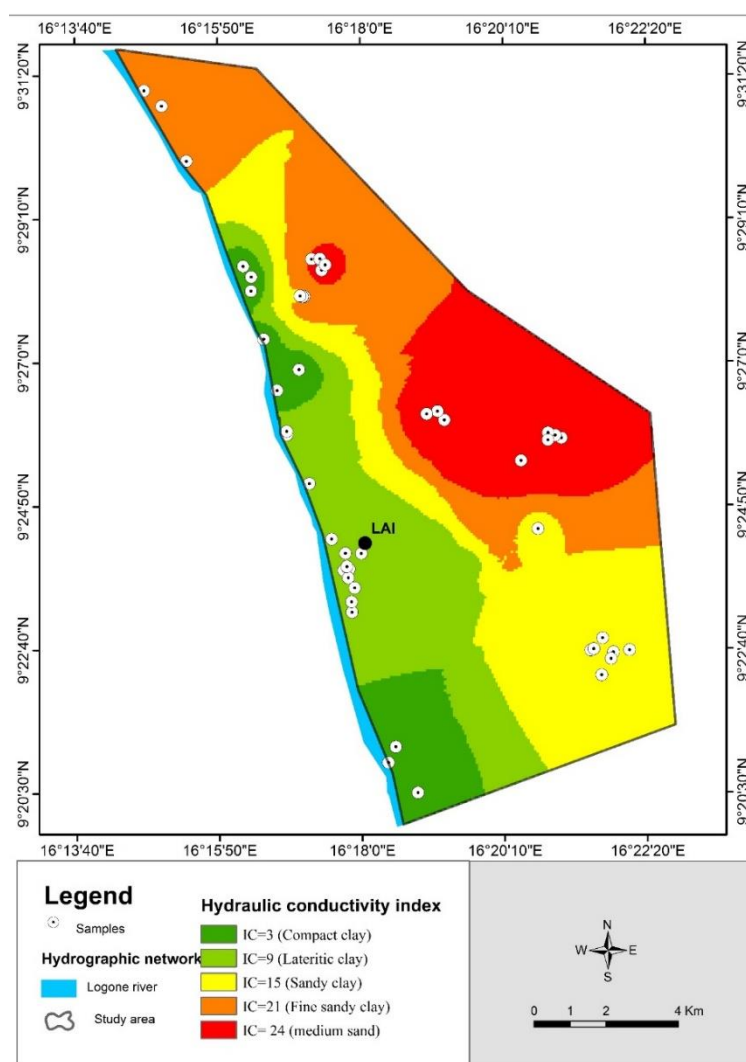


Figure 9. Hydraulic conductivity map.

3.8. DRASTIC Index

The groundwater vulnerability map of Lai City (Figure 10) was generated by calculating the DRASTIC Index (DI) using Eq. (1) through a weighted overlay analysis of seven thematic layers, each representing one of the DRASTIC parameters. The resulting DI values range from 119 to 175 across the study area. Based on the classification proposed by Aller et al. (1987)[7], three groundwater vulnerability classes were identified. Based on the DRASTIC Index (DI) classification, areas with DI values below 119 were categorized as having low groundwater vulnerability, whereas DI values ranging from 119 to 155 and 155 to 175 corresponded to moderate and high vulnerability classes, respectively. The spatial extent of each vulnerability class is presented in Table 3. The northern and southwestern parts of the study area exhibit high groundwater vulnerability to contamination, covering approximately 35% of the total study area (Table 3). These highly vulnerable zones include Lai, Gabri Mbassa, Sategui, and Djongo. Field observations and available soil data indicate that these areas

are predominantly characterized by hydromorphic soils (Figure 6). In addition, the depth to the water table is relatively shallow, ranging from 1.5 to 4.5 m. The combination of shallow groundwater levels and hydromorphic soil conditions contributes to the elevated vulnerability of these areas by favoring infiltration and facilitating the potential transport of contaminants toward the groundwater system. In contrast, the southeastern part of the study area is characterized predominantly by low groundwater vulnerability, accounting for approximately 33% of the total area (Table 3). The greater depth to the water table and other hydrogeological characteristics of this sector likely contribute to its relatively lower susceptibility to contamination. This low-vulnerability sector includes the localities of Djoun, Totalie, and Madjang. Field observations indicate that these areas are predominantly characterized by halomorphic soils and relatively deep groundwater levels, ranging from 4.5 to 9.0 m. These hydrogeological conditions are associated with relatively low groundwater vulnerability in this sector, primarily due to the greater depth to the water table. Bera et al. (2021) [13] noted that contaminants migrating through the vadose zone generally require more time to

reach the groundwater table when groundwater is deeper, allowing greater opportunity for natural attenuation. In contrast, the northeastern and central-eastern sectors of the study area, particularly around Tchindai and Totalie, display low to mod-

erate groundwater vulnerability. This spatial heterogeneity results from the combined effects of groundwater depth and the other hydrogeological parameters integrated into the DRASTIC model.

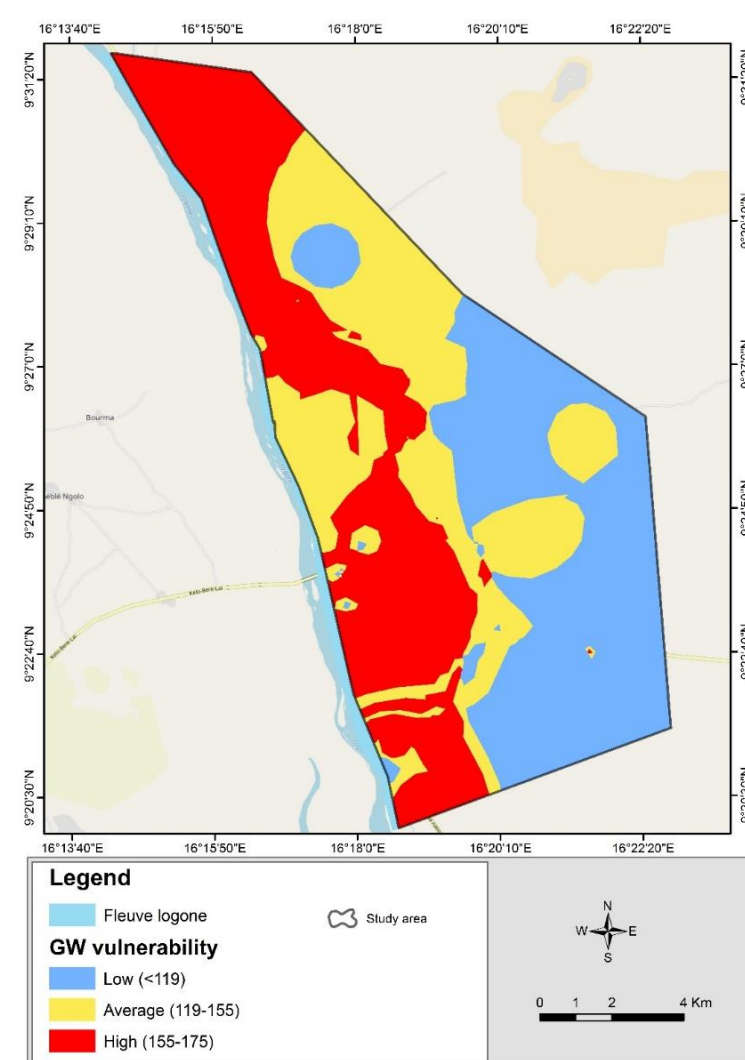


Figure 10. Groundwater vulnerability map of Lai city.

Table 3. DRASTIC index classification of groundwater vulnerability zones.

Groundwater vulnerability class	Drastic index classes	Area (sq.km)	Area (%)	Well locations
Low	< 119	51.98	34	Djoun, Totalie, Madjang
Average	119-155	46.75	31	Tchindai
High	155-175	52.38	35	Lai, Sategui, Djongo

3.9. DRASTIC Index Validation

The reliability of the GIS-based DRASTIC model used to

generate the groundwater vulnerability map (Figure 10) was evaluated using nitrate and Total Dissolved Solids (TDS) concentrations as validation parameters. We collected groundwa-

ter samples from 50 locations across the study area. The spatial distributions of nitrate and TDS concentrations in groundwater are presented in Figs. 11 and 12, respectively. Results of sample analysis indicated that nitrate and TDS values in the study area vary from 0 to 68.2 ppm and 11-518 ppm, respectively. Figure 11 and Figure 12 show clearly that the distribution of the parameters (Nitrate and TDS) in the south-western portion of the basin, namely Lai and Gabri Mbassa, is a high concentration of nitrate, ranging from 40.8 to 68.2 ppm, and a relatively high concentration of TDS, ranging from 266 to 518 ppm. As shown in Figure 10, the Lai and Gabri Mbassa areas are characterized by high groundwater vulnerability. These ar-

eas are located along the Logone River, where intensive irrigated agriculture is practiced. Under these conditions, irrigated agriculture may provide contaminants via drainage waters during the groundwater recharge process. On the other hand, the south eastern sides of the basin show low concentrations of TDS and nitrate, ranging from 0 to 13.4 ppm and 11-102 ppm, respectively. This part of the study area shows lower groundwater vulnerability where Djoun, Totalie, Madjang areas are located. The occurrence of higher nitrate and TDS concentrations in areas classified as highly vulnerable, together with lower concentrations in areas of low vulnerability, provides supporting evidence for the validity of the groundwater vulnerability map [13].

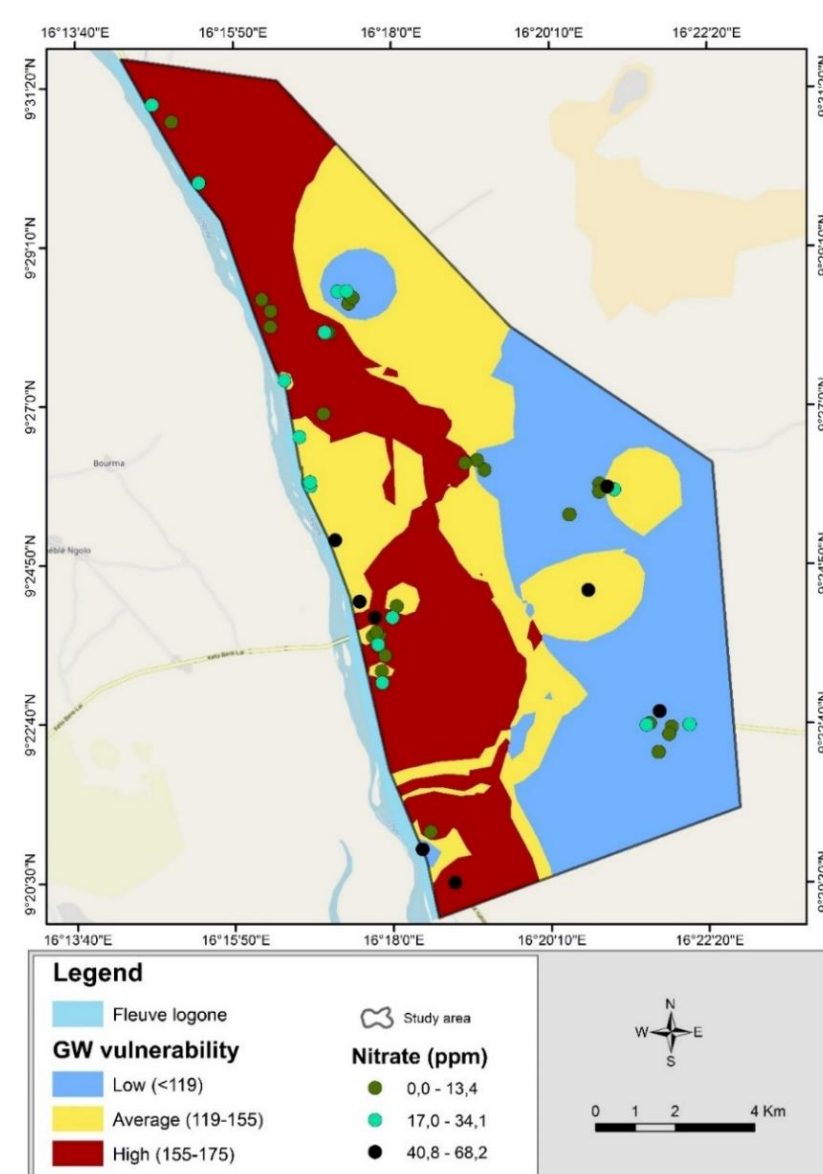


Figure 11. Spatial Distribution of the DRASTIC Index and Nitrate Concentrations.

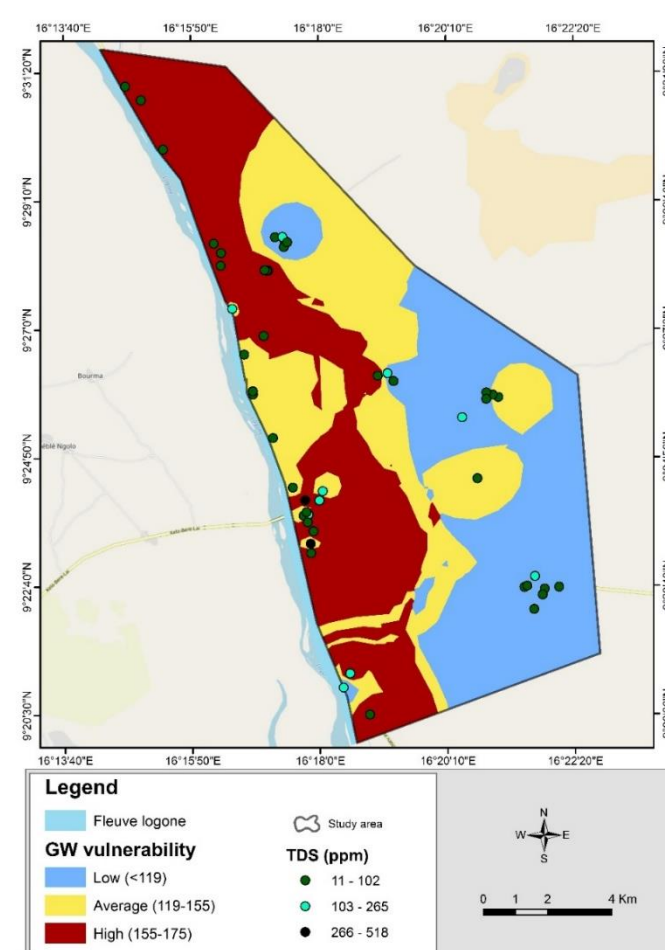


Figure 12. Spatial Distribution of the DRASTIC Index and TDS Concentrations.

The link between groundwater vulnerability, expressed by the DRASTIC Index (DI), and nitrate and TDS concentrations was further examined using correlation analysis (Figures 13 and 14). The results show a strong positive relationship between the aggregated maximum concentrations of both parameters and the DRASTIC Index. The Pearson correlation coefficients between groundwater vulnerability and the maximum concentrations of nitrate and TDS were $r = 0.97$ and $r = 0.75$, respectively, indicating strong positive correlations. The

observed positive correlations, together with the higher concentrations of nitrate and TDS recorded in areas classified as highly vulnerable, provide supporting evidence for the reliability of the groundwater vulnerability map. Similar findings have been reported by El-Meknassi et al. (2021) [20], who demonstrated that the occurrence of elevated concentrations of groundwater quality indicators in highly vulnerable areas can be used to support the validation of vulnerability assessments.

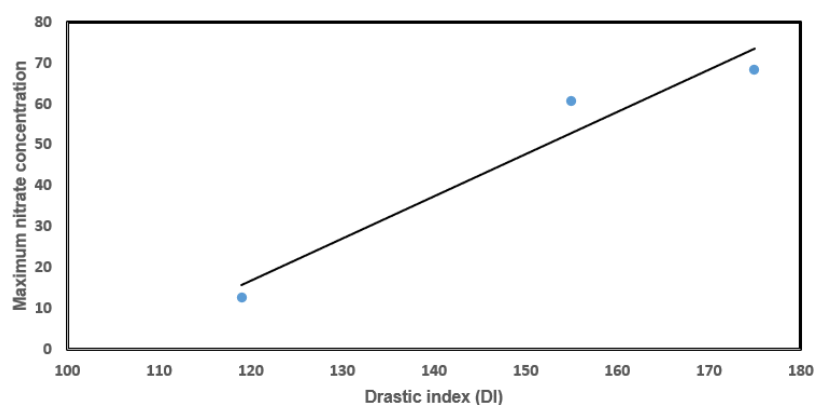


Figure 13. Correlation between DRASTIC Index and maximum nitrate concentrations.

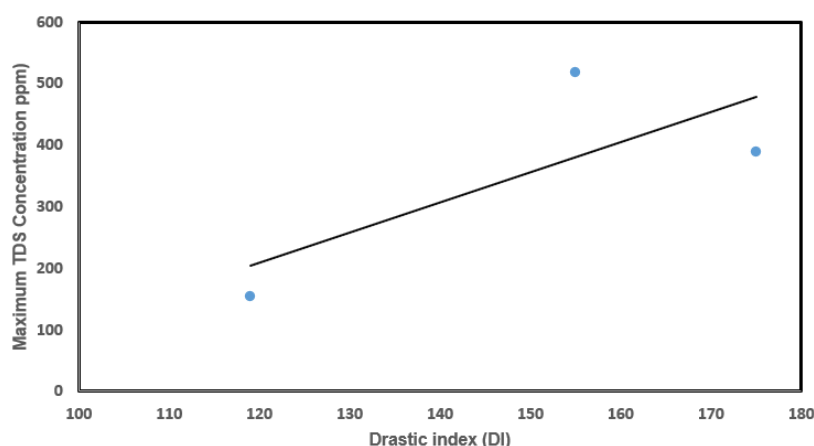


Figure 14. Correlation between DRASTIC Index and maximum nitrate concentrations.

3.10. Discussion

This study represents the first comprehensive effort to assess groundwater contamination and apply a GIS-based DRASTIC vulnerability assessment in southern Chad, with particular emphasis on Lai City. As already mentioned, groundwater vulnerability assessment of Lai city is needed in order to carefully predict pollution that is faced by the city and its surroundings. Based on the nitrate and TDS results presented above, groundwater quality in Lai City is degraded in certain areas, which correspond closely to zones characterized by high groundwater vulnerability.

In this study, the original DRASTIC model developed by Aller et al. (1987) [7] was applied to assess the intrinsic vulnerability of groundwater in Lai City. The resulting DRASTIC Index (DI) values indicated a spatial distribution ranging from low to high vulnerability. The groundwater vulnerability map revealed that areas characterized by shallow groundwater levels (1.5–4.0 m), particularly Lai, Gabri Mbassa, Sategui, and Djongo, exhibit the highest degree of vulnerability. This finding is consistent with previous studies by Bojórquez-Tapia et al. (2009) [15], Bera et al. (2021) [13], and El-Meknassi et al. (2021) [20], which identified shallow groundwater levels as a key factor contributing to increased groundwater vulnerability. The high vulnerability observed in these areas can be attributed primarily to the shallow groundwater table and gentle slopes (0–2%), which favor infiltration and facilitate the downward migration of potential contaminants. In addition, agricultural activities in parts of the study area, particularly around Gabri Mbassa and Sategui, may further increase the potential for contaminant infiltration into the groundwater system. These findings are consistent with recent studies by Said et al. (2025) [56] and Oyda et al. (2025) [48], which reported similar groundwater vulnerability patterns using the DRASTIC approach in comparable hydrogeological settings.

The accuracy of GIS-based Drastic model have been evaluated through nitrate and TDS parameters. As mentioned ear-

lier, validation of the Drastic model from observed contaminants datasets throughout the study area, such as nitrate and/or TDS is commonly applied by different authors [13, 20, 48, 56]. We found that elevated nitrate levels (> 45 mg/l) align with areas of high vulnerability, a correlation consistent with other studies [13, 20, 48, 56]. The primary contamination sources are agricultural inputs (e.g., fertilizers) because in the areas of high vulnerability (Lai and Gabri Mbassa) located along the Logone River, intensive irrigated agriculture is practiced. Conversely, the south eastern sides of the study area where Djoun, Totalie, Madjang areas are located, show low concentrations of TDS and nitrate. This is attributed to minimal agricultural activity, resulting from poor soils (halomorphic soil) unsuitable for farming. This finding is consistent with those obtained by Oyda et al. (2025) in similar areas.

4. Conclusion

The present study assessed groundwater vulnerability in Lai City, located in the Tandjile-East Department of southern Chad, using the GIS-based DRASTIC model and seven hydrogeological factors are considered. The resulting DRASTIC Index (DI) values ranged from 119 to 175, allowing the study area to be organized into three groundwater vulnerability classes: low, moderate, and high. The high-vulnerability class accounted for approximately 35% of the study area, while the moderate- and low-vulnerability classes covered approximately 31% and 34%, respectively. The reliability of the GIS-based DRASTIC assessment was further evaluated using nitrate and Total Dissolved Solids (TDS) concentrations. Higher nitrate and TDS concentrations were generally associated with areas classified as highly vulnerable, whereas lower concentrations occurred predominantly in areas of low vulnerability. The Pearson correlation coefficients between groundwater vulnerability and the maximum concentrations of nitrate and TDS were $r = 0.97$ and $r = 0.75$, respectively, indicating strong positive relationships. These results provide supporting evidence for the reliability of the GIS-based DRASTIC vulnerability assessment. The northern and southwestern parts of the

study area were identified as zones of high groundwater vulnerability. These areas are characterized by shallow groundwater levels and predominantly hydromorphic soils, conditions that favor infiltration and the downward transport of potential contaminants. The most vulnerable areas, including Lai, Gabri Mbassa, Sategui, and Djongo, are located along the Logone River, where intensive irrigated agricultural activities may further contribute to the risk of groundwater contamination. Overall, the groundwater vulnerability map developed in this study provides a useful tool for groundwater resource protection, contamination risk management, and land-use planning in Lai City and its surrounding areas.

Author Contributions

Tidjani Bahar: Conceptualization, Investigation, Methodology, Resources, Supervision, Writing – original draft, Writing – review & editing

Mahamat Moussa: Conceptualization, Data curation, Investigation, Software

Conflicts of Interest

The authors declare that they have no competing interests.

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